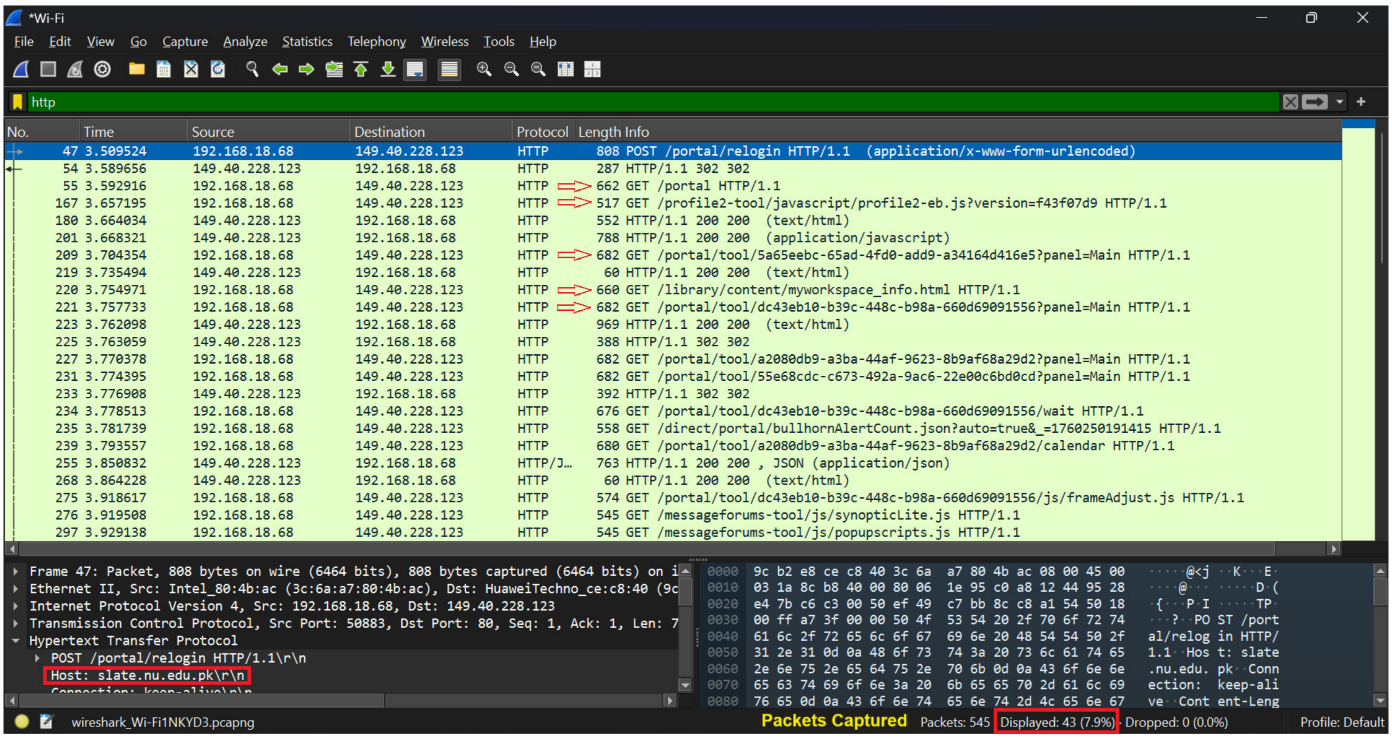
CNET Assignment 1

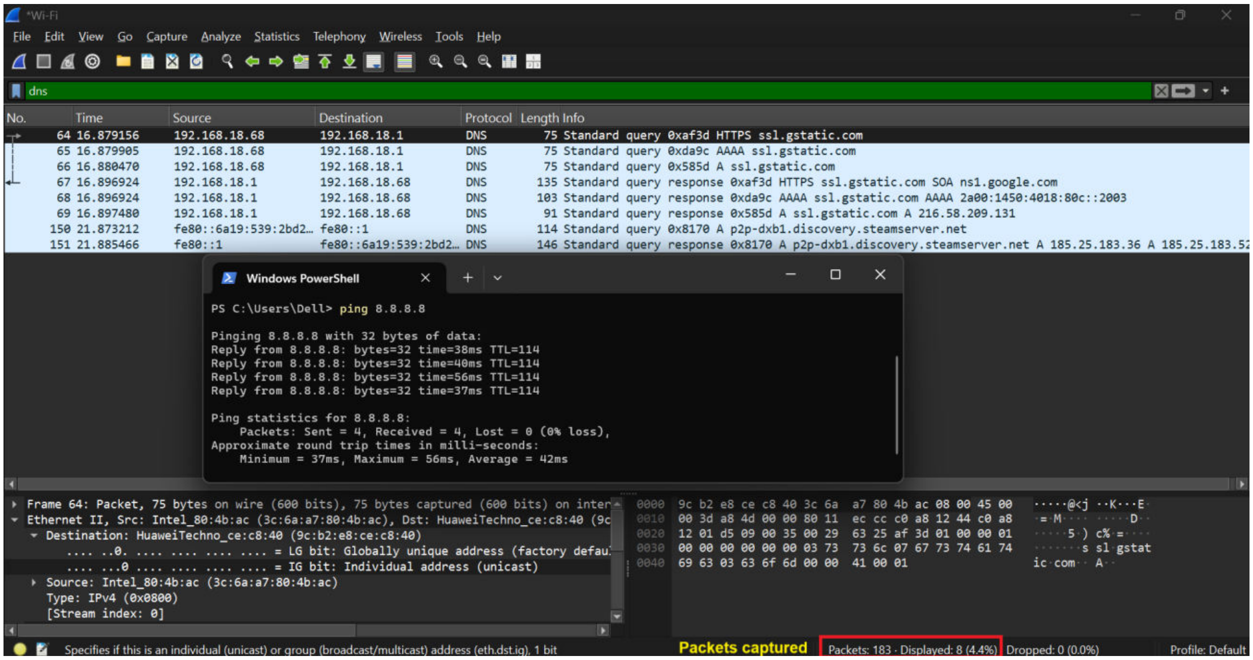
Question 3

1. Capture Live HTTP, FTP, and DNS Packets (at least 5 packets)

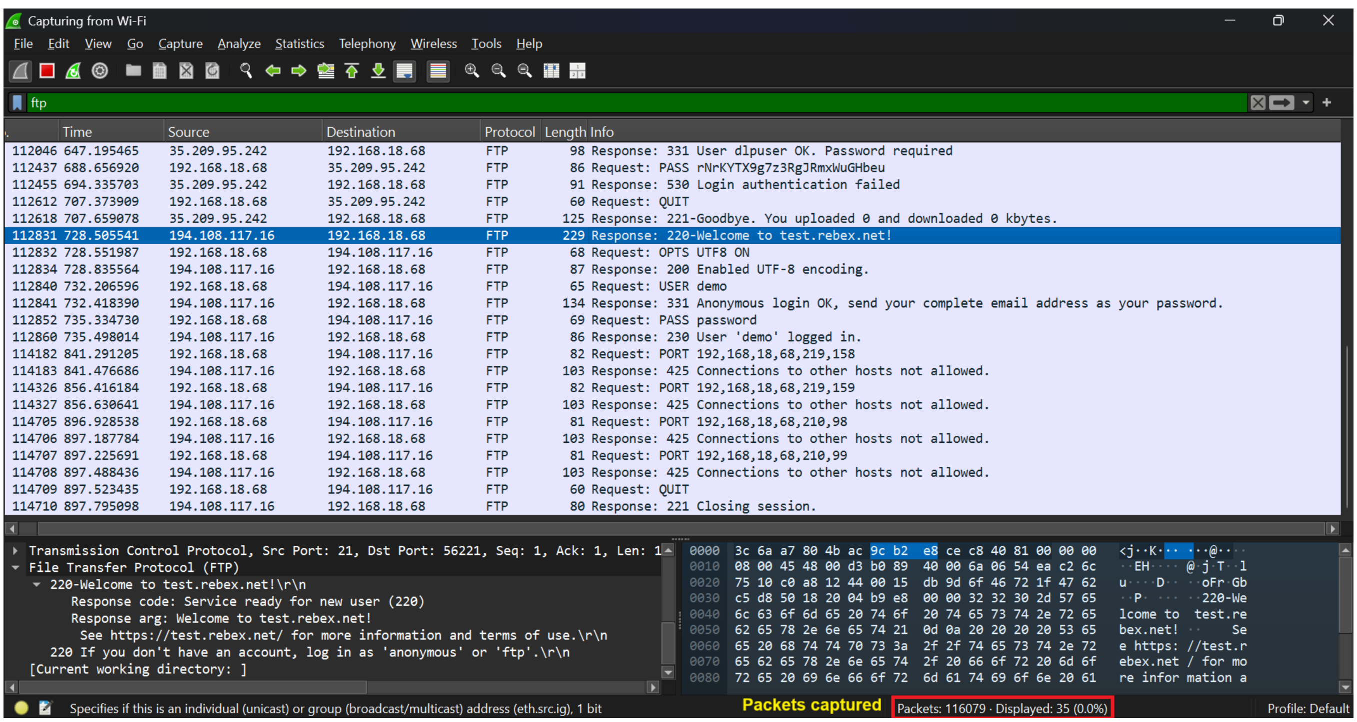
- **Use Wireshark to filter and capture packets for:**
  - **HTTP GET requests**



  - **DNS Query responses**

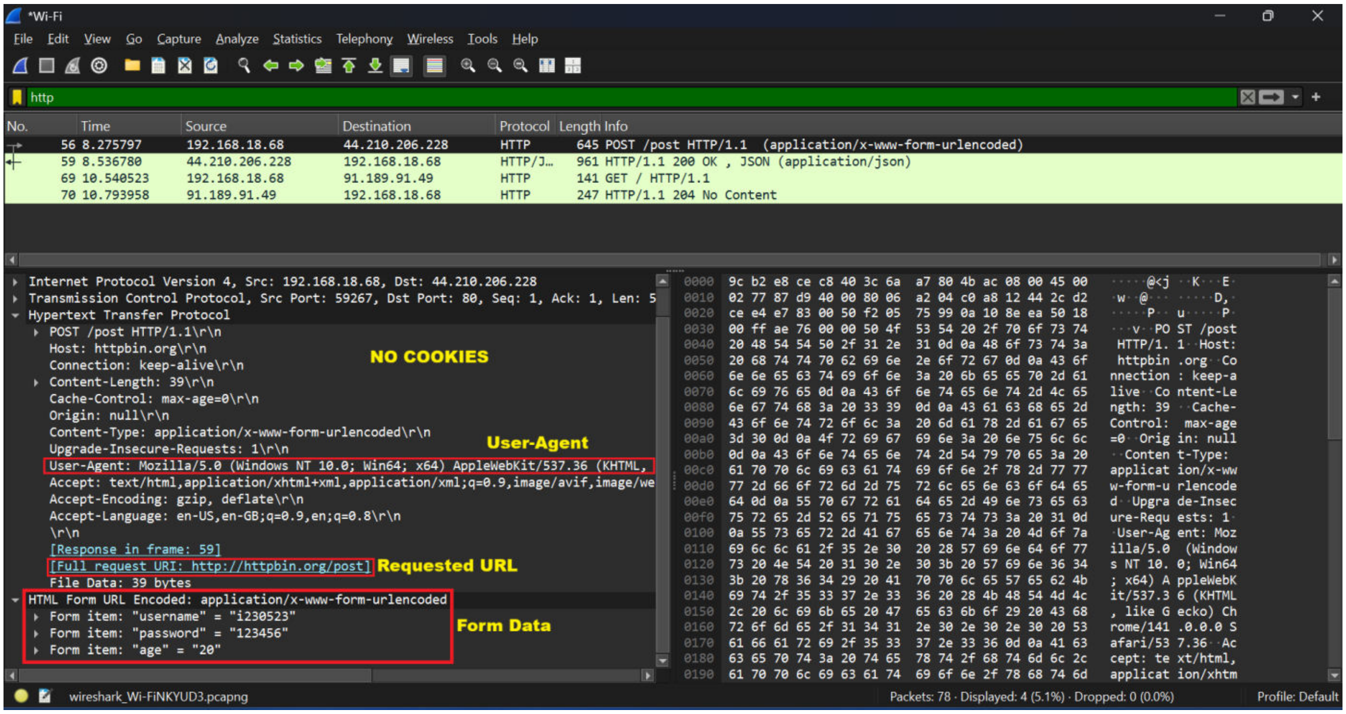


- **FTP file transfer data**



2. Analyze a captured HTTP request

- **Identify the requested URL, User-Agent, and cookies.**



- **Explain how HTTPS protects sensitive data compared to HTTP.**

Ans: HTTPS uses TLS/SSL encryption to secure communication between client and server.

The HTTPS:

- Encrypts all the form data such as username, password, etc.
- Authenticates server's identity through digital certificates.
- Prevents attackers from modifying information during data transmission.

Q4

(i)

Transmission Rate $(R) = 350$ Mb/s
Object size $(L) = 250,000$ bits
Request Rate $(\lambda) = 35$ req/s
Avg internet-side round trip $(T_{internet}) = 2.5$ sec
(i) Total avg response time $(T_{total}) = ?$

Data rate (Arrival rate) $= 250\,000 * 35$
$= 8.75 \times 10^6$ bits/s
$= 8.75$ Mb/s

$$\text{Utilization}(\rho) = \frac{\text{Arrival rate}}{\text{Transmission rate}}$$

$$= \frac{8.75 \text{ Mb/s}}{350 \text{ Mb/s}}$$

$= 0.025$

$= 2.5\%$

$$\text{Access delay } (T_{access}) = \frac{L/R}{1-\rho}$$

$$= \frac{250000/(350 \times 10^6)}{1 - 0.025}$$

$$= 0.73 \times 10^{-3}$$

$\because T_{total} = T_{access} + T_{internet}$

$= (0.73 \times 10^{-3}) + 2.5$

$= 2.50073$

$\boxed{T_{total} \approx 2.5 \text{ sec}}$ Ans.

(ii) .

Miss rate $= 0.7$
Response time of cache $= 0.01$

$\Rightarrow$ Hit rate $= 1 - 0.7$
$= 0.3$

$\Rightarrow$ Total response time (with cache):

$$T_{total(c)} = (\text{Miss rate} * T_{total}) + (\text{HitRate} * \text{Response cache time})$$

$$= (0.7 * 2.50073) + (0.3 * 0.01)$$

$\boxed{T_{total(c)} = 1.753 \text{ sec}}$ Ans.

Q5

Data:

Buffer max capacity $(b_{max}) = 50\,000$ packets
Packet size $(P_{size}) = 1000$ bytes
Ongoing link $= 100$ Mbps
Arrival link $= 150$ Mbps
$t = 10s$

1- Time for buffer overflow = ?
2- Packets dropped = ?
3- Advanced mechanisms (2) = ?

Sol:

$\because$ 1 byte = 8 bits

$$b_{max} = 50\,000 \times (1000 \text{ bytes}) \times 8 \text{ bits}$$
$$= 400 \times 10^6 \text{ bits}$$
$$= 400 \text{ M bits}$$

Buffer overload = (Arrival link) − (Ongoing link)
" $= 150 - 100$
$V_{overload} = 50$ Mbps

Overloaded data in t secs = $b_{overload} \times t$

$= (50 \times 10)$ (Mbps*s)

$= 500$ Mbits

Time for buffer overflow $= \dfrac{b_{max}}{V_{overload}}$

$= \dfrac{400 \text{ Mbits}}{50 \text{ Mbps}}$

1. $\boxed{t_{overflow} = 8 \text{ s}} \rightarrow$ Ans.

Data dropped $= (500 - 400)$ Mbits

$= 100 \times 10^6$ bits

$= (100 \times 10^6) \div 8$

$= 12.5 \times 10^6$ bytes

Packets dropped $= (12.5 \times 10^6) \div 1000$

2. $\boxed{= 12500 \text{ packets}}$ Ans.

3_ Advanced Mechanisms:

① Priority Queueing:

- Router maintains multiple queues.
- Different traffic goes into different queues.
- High priority packets gets transmitted.
- When buffer fills, it drops low priority packets, first.

② Congestion Window Adjustment (At TCP Level):

- TCP sender maintains a congestion window (packets it can send)
- Starts small, then gradually increase.
- If packet loss detected, it immediately reduce window size.
- If no loss occurs, it gradually increases again.